File permissions in Linux

Project description

In this activity, I used Linux Commands to review and manage file and directory permissions. I checked existing permissions, identified files with incorrect access settings, and modified them using the chmod command. I also worked with hidden files and directories to ensure that only authorized users had access.

Check file and directory details

Command used:
ls -la
The ls -la command was used to display all files and directories, including hidden files. This command showed the permission strings, owners, groups, and other details needed to identify unauthorized access and determine which permissions needed to be modified.

Describe the permissions string

Example permission string: -rw-rw-r—
The first character (-) indicates that the item is a regular file. The next three characters (rw-) represent the permissions for the file owner, meaning the owner can read and write the file. The following three characters (rw-) represent permissions for the group. The last three characters (r–) represent permissions for others. A dash (-) indicates that a permission is not granted.

Change file permissions

Command used:
chmod o-w project_k.txt
This command removed write permissions for others on the project_k.txt file. The organization's policy does not allow others to have write access to files, so this permission was removed to improve security.

Change file permissions on a hidden file

Command used:
chmod u-w,g-w .project_x.txt

This command removed write permissions from the user and group for the hidden file .project_x.txt. After the change, the user and the group retained read access while access was removed, which matched the required authorization settings.

Change directory permissions

Command used:
chmod g-x drafts
This command removed execute permissions from the group for the drafts directory. As a result, only the file owner could access the directory and its contents, preventing unauthorized access by group members.

Summary

This activity gave me hands-on experience working with linux file directory permissions. I used linux commands such as ls -la and chmod to review existing permissions and make changes where access was not aligned with security requirements. I also worked with hidden files and directories to ensure that only authorized users could access sensitive information. Through this activity, I gained a better understanding of how linux permissions help maintain system security.